

LAITH ASSAF

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SUPERSET ARCHIVE – assemble one-page variants from this file; do not submit it

EDUCATION

Michigan State University, College of Engineering

East Lansing, MI

BS Computer Science, AI/ML | Minor in Economics | Dean's List x3

August 2022 - May 2027

- [SWE] **Coursework:** Data Structures & Algorithms (CSE 331), Software Design (CSE 335), Computer Systems (CSE 325), Computer Organization & ARM Assembly (CSE 320)
- [AI] **Coursework:** Data Structures & Algorithms (CSE 331), Cloud Computing & Database Systems (CSE 380), Software Design (CSE 335), Computer Organization (CSE 320)
- [IT/SA] **Coursework:** Cloud Computing & Database Systems (CSE 380), Computer Systems (CSE 325), Computer Organization & ARM Assembly (CSE 320), International Trade Economics (EC 440)

EXPERIENCE

PC Technician Intern

May 2026 - Present

Onsite Support Team, Delta Dental

Okemos, MI

- [IT/SA] Deployed and configured **150+** workstations (monitors, docking stations, keyboards, mice) across a new corporate office buildout, imaging endpoints via **Ivanti** MDM to enforce standardized enterprise configuration
- [SWE/IT] Developing an automated ticket creation workflow in **Power Automate** to streamline IT ops; performed PC imaging and disk backups and drive kills across endpoint lifecycle tasks

IT and Media Services Support Associate

May 2023 - Present

MSU College of Osteopathic Medicine

East Lansing, MI

- [IT/SA] Configured and maintained 100+ classroom technology systems (projectors, audio systems, video conferencing) across 4 locations supporting **1,200+ students**, resolving 50-70 technical tickets weekly with 5-10 minute average response time
- [IT] Engineered a systematic troubleshooting framework maintaining **99.5% classroom uptime** across live lectures, diagnosing network connectivity, display output, audio routing, and Zoom conferencing failures within a 5-minute SLA
- [SWE/AI] Supported 100+ classroom AV and conferencing systems across 4 buildings serving **1,200+ students**, resolving 50-70 tickets weekly and maintaining **99.5% uptime** through a documented diagnostic methodology

PROJECTS

[AI/SWE] **Pokemon Battle Agent - Reinforcement Learning** | *PyTorch, PPO, MCTS*

May 2026 – Present

- Trained a competitive battle agent through **behavior cloning** on human replay data, **anchored PPO** fine-tuning of an MLP policy/value network over structured per-Pokemon token observations, and **determinized MCTS** at decision time, reaching **93% win rate** against scripted baselines
- Built a parallel battle-simulation and evaluation harness on the Pokemon Showdown engine, running **747 rated ladder games** across **11 milestones** against a pre-registered evaluation methodology written before each experiment
- Ran a code review that uncovered **four confounds** invalidating prior results (greedy vs. stochastic decoding accounting for the entire reported search gain; opponent concessions inflating ladder win rate by **~7pp**), then restated every affected number and reopened two milestones rather than reporting the inflated figure
- Tested and retired a transformer policy against an MLP baseline, documenting six hypotheses returned null or weak, to isolate observation quality from search as the binding constraint

[AI/SWE] **Agentic Development System** | *Claude Code, subagents, hooks, MCP*

January 2026 – Present

- Designed a **10-agent orchestration pipeline** (planner → parallel builders → reviewer, tester, simplifier, docs auditor) with a single transient plan file as the handoff contract, applied across **44 repositories**
- Instrumented **11 lifecycle hook events** (tool pre/post, tool failure, subagent start/stop, session start/end) to enforce project conventions automatically instead of relying on prompt discipline
- Authored a tiered documentation protocol with **pre-registered success gates**, requiring every milestone to define a testable criterion ("LCP < 2.5s measured by X") before implementation begins, plus **5 reusable skills** for debugging, adversarial review, and requirements elicitation

- Wrote **6 custom agents** and cross-repository index documents coordinating a multi-package monorepo, cutting context re-derivation between sessions

[AI/SWE] **SoundSense - Real-Time Audio Classification** | *YAMNet, Flask* | **2nd Place, MHacks** October 2025

- Placed **2nd of 300+ participants at MHacks 2025** (24-hour hackathon) building a real-time audio classification system using **YAMNet** CNN on MemryX MXA2301 edge accelerators, achieving **95% accuracy** at <50ms latency across 521 AudioSet classes
- Delivered hazard alerts through a **Flask + WebSocket** pipeline to a live web interface for hearing-impaired users, running inference fully on-device

[SWE] **Bonfire Platform & Ember Framework** | *TypeScript, Socket.io, Next.js* December 2025 – April 2026

- Designed and published **Ember**, an open-source TypeScript framework for multiplayer party games (@**bonfire-ember/core/server/client** npm packages), implementing phase state machine, WebSocket room orchestration, and player lifecycle into a 5-hook API enabling **80%+ code reuse** across new game builds
- Architected a **multi-room orchestration system** on Express + **Socket.io** namespaces with **Firestore Realtime DB** state sync, hosting 4 concurrent games on a single **Railway** server with configurable disconnect strategies and TTL-based cleanup
- Shipped **64 commits** across three packages (framework, platform, games) with a documented issue-flow contract between the framework repo and its consumer game repos

[AI/SA] **AI-Assisted Comparative Market Analysis** | *Python, scikit-learn, Streamlit* June 2026 – Present

- Built a sold-price prediction model over metro Detroit **MLS** history achieving **13.3% MAPE** with **52% of estimates within 10%**, surfacing the comparable sales behind each estimate so the output is auditable rather than a bare number
- Diagnosed the accuracy ceiling as a data problem, not a model problem: square footage present on only **17%** of sold records and lot size, parking, and condition missing entirely; documented the gap against the **5% MAPE** agent-grade bar and gated the product roadmap on closing it instead of shipping

[AI/SWE/IT] **Nexus - AI-Powered SSH Terminal Management** | *FastAPI, Svelte, Ollama, Docker* 2025

- Built a multi-server SSH management platform pairing live terminal access with a **locally hosted LLM** (Ollama, gpt-oss:20b), streaming natural-language requests into suggested shell commands behind a **safety validation** layer so no generated command executes unreviewed
- Implemented real-time terminal I/O over **WebSockets** with a **FastAPI** backend and Svelte 5 frontend, handling multi-server connection state and session lifecycle; containerized with **Docker**

[AI/SA] **Toyota GR Cup Telemetry Analysis** | *Python, PostgreSQL, scikit-learn* 2025

- Modeled tire degradation across **3,257 laps** from 7 racetracks at a hackathon, reaching **$R^2 = 0.631$** on high-frequency telemetry loaded into **PostgreSQL** with ML-ready preprocessing pipelines

[IT/SA] **Self-Hosted Home Lab Infrastructure** | *Linux, Docker, Networking* 2023 – Present

- Operate a rack server (**128GB+ RAM**) running containerized media, game, and **VPN** services under **Docker**, self-administering Linux, networking, and service orchestration as a production-like environment

[SWE] **Personal Portfolio Site** | *Next.js, TypeScript, Tailwind* 2026

- Built and deployed a personal portfolio and resume site in **Next.js + TypeScript**, shipped across **34 commits**

LEADERSHIP

Michigan State University | AI Club East Lansing, MI
Workshop Coordinator January 2025 - Present

- [AI/SWE] Developed and delivered 8+ technical workshops on **local LLM deployment** (Ollama, LM Studio), AI ethics, and ML fundamentals (**PyTorch**) to 500+ attendees across 15 sessions, recording 90% retention and 4.8/5 satisfaction score
- [AI] Built and taught a **multi-agent orchestrator** demo coordinating Scope, Architecture, Builder, and Reviewer agents to plan and generate a web application from a one-line idea, used as live workshop material

Michigan State University | AI Club East Lansing, MI
Project Lead January 2025 - May 2025

- [AI/SWE] Led a 4-student team building an Academic Performance Predictor on the **GPT-4 API** and prompt engineering, shipping course-level predictions across 50+ courses on agile sprints

- [MERGED] Delivered 8+ technical workshops on **local LLM deployment** (Ollama, LM Studio), agent orchestration, and **PyTorch** fundamentals to 500+ attendees, and led a 4-student team shipping an **LLM-powered** academic performance predictor across 50+ courses

TECHNICAL SKILLS

Languages: Python, C++, C#, TypeScript, JavaScript, SQL, ARM Assembly

AI/ML: PyTorch, scikit-learn, pandas, NumPy, PPO / reinforcement learning, MCTS, behavior cloning, CNNs

AI Engineering: LLM APIs (Anthropic, OpenAI), agent orchestration, MCP, prompt engineering, local inference (Ollama, LM Studio), model evaluation

Web & Backend: FastAPI, Flask, Express, Node.js, React, Next.js, Socket.io, REST APIs, WebSockets, Streamlit

Data & Infrastructure: PostgreSQL, Firebase, Docker, Linux, Git, Railway, Vercel, networking / VPN

IT & Enterprise: Active Directory, Intune, Ivanti MDM, Power Automate, endpoint imaging, ticketing workflows